

Andrew_Crawford_Lab4

Packages for cleaning messy data

```
library(stringr)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
headers_fasta <- readLines("messy_sequences.fasta")
headers_fasta <- headers_fasta[str_starts(headers_fasta, ">")]
headers_fasta
```

```
## [1] ">sample_001|organism=Homo_sapiens|gene=BRCA1|len=120"
## [2] ">Sample002 organism:Homo sapiens gene:TP53"
## [3] ">sample-003 | Homo_sapiens | EGFR | length=150bp"
## [4] ">SAMPLE_004;species=H.sapiens;target=BRCA1"
## [5] ">sample005 Homo sapiens; TP53; 130 bp"
## [6] ">seq6|Hsapiens|EGFR"
## [7] ">Sample_007 organism=Homo_sapiens|gene=BRCA1|note:re-sequenced"
## [8] ">sample-8 species:Homo_sapiens gene:TP53 len:NA"
```

Cleaning messy_sequence.fasta

```

sample_id_pattern <- "((?i)>s[\\w_-]+)"
organism_pattern <- "(H\\.\\.\\. [\\w_-]+)"
gene_name_pattern <- "(gene.(\\w+)|(target.(\\w+)|(EGFR)|(TP53))"
gene_len_pattern <- "(( \\d\\.+)|(len.+))"

extracted <- data.frame(
  sample_id = str_extract(headers_fasta, sample_id_pattern),
  organism = str_extract(headers_fasta, organism_pattern),
  gene_name = str_extract(headers_fasta, gene_name_pattern),
  gene_len = str_extract(headers_fasta, gene_len_pattern),
  stringsAsFactors = FALSE
)

clean_sequence <- extracted

View(clean_sequence)

write.csv(clean_sequence, "clean_sequence.fasta")

```

Cleaning messy_samples.csv

```

messy_sample <- read.csv("messy_samples.csv")
messy_sample

```

##	sample_id	patient_name	dob	sex	enrollment_site	glucose_value
## 1	S0001	A. Nguyen	07/09/1962	F	Site A	75.4
## 2	S0002	T. Patel	12/16/1974	Male	SITE-A	120.0
## 3	s-0003	R. Martinez	11.24.53	F	Site B	84.2
## 4	S0004	B. Smith	12/18/1975	Female	Site B	120.0
## 5	S0005	B. Martinez	1997-12-02	Male	SITE-A	248.1
## 6	S0006	A. Johnson	1955-12-09	f	Site B	141.2
## 7	S0007	T. Martinez	1969-09-04	unknown	Site A	225.0
## 8	s-0008	T. Martinez	21-Jun-1997	M	Site A	192.7
## 9	S0009	A. Nguyen	2018-05-11	F	SITE-A	221.5
## 10	S0010	A. Martinez	10/01/1951	f	Site C	215.2
## 11	s-0011	K. Garcia	1997-10-16	Male	siteB	99.7
## 12	S0012	S. Kim	21-Apr-2008	unknown	Site B	N/A
## 13	S0013	M. Smith	08/14/2006		Site A	224.7
## 14	S0014	K. Patel	1986-07-09	F	Site B	133.8
## 15	S0015	B. Smith	06/10/1980	Female	siteB	150.8
## 16	s-0016	B. Smith	07/01/1998	F	Site C	112.5
## 17	s-0017	T. Smith	10.17.73	m	Site C	249.2
## 18	S0018	A. Johnson	05.30.02		siteB	84.4
## 19	S0019	M. Martinez	07/26/2005	U	siteB	164.1
## 20	S0020	S. Brown	06/15/2017	M	siteB	249.3
## 21	S0021	B. Patel	02.26.64	m	Site C	187.7
## 22	S0022	J. Nguyen	1959-08-09	m	site_a	199.2
## 23	S0023	A. Garcia	10/28/1974	M	site_a	184.7
## 24	S0024	B. Garcia	2011-12-30	M	siteB	169.5
## 25	S0025	A. Johnson	05/21/1972	f	Site C	143.1

## 26	S0026	J. KIM	30-Jan-2016	U	Site C	202.3
## 27	S0027	K. Patel	19-Apr-1991	m	Site C	111.0
## 28	s-0028	S. Johnson	09-Nov-1985	M	SITE-A	102.1
## 29	S0029	T. Martinez	05.17.60	Female	Site C	148.5
## 30	S0030	R. Garcia	10-Dec-1976	U	Site C	123.2
## 31	S0031	J. Martinez	06/07/1986	m	Site B	211.4
## 32	S0032	J. Garcia	23-Jul-2009	F	Site C	112.3*
## 33	S0033	K. Martinez	07.14.56	U	siteB	105.0
## 34	s-0034	B. Kim	06.07.12	M	Site C	91.8
## 35	S0035	K. Martinez	04.15.95	unknown	Site C	225.1
## 36	S0036	K. Patel	13-Dec-1956	Female	SITE-A	230.7
## 37	S0037	B. Kim	09.06.98		Site C	145.6
## 38	s-0038	R. Nguyen	1987-08-05	unknown	Site B	157.4
## 39	S0039	K. Lee	12/02/1997	U	Site C	74.9
## 40	S0040	M. Brown	04.17.18	m	Site C	145.9
## 41	S0041	M. Garcia	06/21/2008	M	site_a	221.1
## 42	S0042	K. Nguyen	01/20/1965	Male	Site A	174.2
## 43	S0043	M. Garcia	2006-04-10	F	SITE-A	223.1*
## 44	S0044	R. Garcia	08/12/1980	U	Site A	148.0
## 45	S0045	S. Martinez	04.13.98	unknown	Site C	176.6
## 46	S0046	M. Brown	12.12.09		Site C	128.5
## 47	S0047	S. DAVIS	02/21/1967	Female	Site B	169.9
## 48	S0048	B. Nguyen	2012-01-24	m	SITE-A	155.2
## 49	S0049	S. Nguyen	1960-10-28	f	siteB	241.2
## 50	S0050	S. Garcia	1976-07-30	f	Site A	197.4
## 51	S0051	J. Brown	06-Feb-1951	unknown	SITE-A	103.2
## 52	s-0052	K. Garcia	08/02/2011	Male	siteB	240.8
## 53	S0053	M. Davis	20-Sep-1989	U	siteB	80.8
## 54	S0054	J. Patel	1956-09-25	M	Site C	176.9
## 55	S0055	B. Patel	2006-01-28	U	site_a	186.6
## 56	S0056	S. Brown	14-Oct-2011	U	siteB	N/A
## 57	S0057	R. Martinez	1958-03-12	M	Site C	217.9
## 58	S0058	S. Lee	02.08.89	F	Site B	130.0
## 59	S0059	T. DAVIS	25-Nov-1957	m	Site C	142.8
## 60	S0060	R. Martinez	02.09.06	M	siteB	93.6
##	glucose_unit		notes			
## 1	mg/dl					
## 2	mg/dl					
## 3	MG/DL					
## 4	MG/DL					
## 5	mg/dl					
## 6	mmol/L					
## 7	mg/dl					
## 8	MG/DL					
## 9	mg/dl					
## 10	mg/dl	re-draw requested				
## 11	mmol/L					
## 12	mg/dl	re-draw requested				
## 13	mg/dL					
## 14	mmol/L					
## 15	mmol/L					
## 16	mmol/L					
## 17	mmol/L					
## 18	mg/dl	re-draw requested				

```

## 19      MG/DL
## 20      mg/dL
## 21      mg/dL
## 22      mg/dl
## 23      MG/DL
## 24      mmol/L
## 25      mg/dl
## 26      mg/dl
## 27      mg/dL
## 28      MG/DL re-draw requested
## 29      mg/dl
## 30      mg/dl
## 31      MG/DL
## 32      MG/DL
## 33      mmol/L
## 34      mg/dl
## 35      mmol/L
## 36      mg/dL
## 37      mg/dL
## 38      mmol/L
## 39      mmol/L
## 40      mg/dL
## 41      mg/dl
## 42      MG/DL
## 43      mg/dL
## 44      MG/DL
## 45      MG/DL
## 46      mmol/L
## 47      mg/dl
## 48      MG/DL
## 49      mg/dl
## 50      mg/dl
## 51      mg/dL
## 52      mg/dL
## 53      mg/dL
## 54      MG/DL re-draw requested
## 55      MG/DL
## 56      mmol/L
## 57      mg/dL
## 58      mg/dL
## 59      mg/dL
## 60      MG/DL

```

```
str(messy_sample)
```

```

## 'data.frame': 60 obs. of 8 variables:
## $ sample_id : chr "S0001" "S0002" "s-0003" "S0004" ...
## $ patient_name : chr "A. Nguyen" "T. Patel" "R. Martinez" "B. Smith" ...
## $ dob : chr "07/09/1962" "12/16/1974" "11.24.53" "12/18/1975" ...
## $ sex : chr "F" "Male" "F" "Female" ...
## $ enrollment_site: chr "Site A" "SITE-A" "Site B" "Site B" ...
## $ glucose_value : chr "75.4" "120.0" "84.2" "120.0" ...
## $ glucose_unit : chr "mg/dl" "mg/dl" "MG/DL" "MG/DL" ...
## $ notes : chr "" "" "" "" ...

```

#Sample_Id Standardization

```
sample_id_clean <- str_to_upper(messy_sample$sample_id)
sample_id_clean <- str_remove(sample_id_clean, "-")
messy_sample$sample_id <- sample_id_clean
```

messy_sample

##	sample_id	patient_name	dob	sex	enrollment_site	glucose_value
## 1	S0001	A. Nguyen	07/09/1962	F	Site A	75.4
## 2	S0002	T. Patel	12/16/1974	Male	SITE-A	120.0
## 3	S0003	R. Martinez	11.24.53	F	Site B	84.2
## 4	S0004	B. Smith	12/18/1975	Female	Site B	120.0
## 5	S0005	B. Martinez	1997-12-02	Male	SITE-A	248.1
## 6	S0006	A. Johnson	1955-12-09	f	Site B	141.2
## 7	S0007	T. Martinez	1969-09-04	unknown	Site A	225.0
## 8	S0008	T. Martinez	21-Jun-1997	M	Site A	192.7
## 9	S0009	A. Nguyen	2018-05-11	F	SITE-A	221.5
## 10	S0010	A. Martinez	10/01/1951	f	Site C	215.2
## 11	S0011	K. Garcia	1997-10-16	Male	siteB	99.7
## 12	S0012	S. Kim	21-Apr-2008	unknown	Site B	N/A
## 13	S0013	M. Smith	08/14/2006		Site A	224.7
## 14	S0014	K. Patel	1986-07-09	F	Site B	133.8
## 15	S0015	B. Smith	06/10/1980	Female	siteB	150.8
## 16	S0016	B. Smith	07/01/1998	F	Site C	112.5
## 17	S0017	T. Smith	10.17.73	m	Site C	249.2
## 18	S0018	A. Johnson	05.30.02		siteB	84.4
## 19	S0019	M. Martinez	07/26/2005	U	siteB	164.1
## 20	S0020	S. Brown	06/15/2017	M	siteB	249.3
## 21	S0021	B. Patel	02.26.64	m	Site C	187.7
## 22	S0022	J. Nguyen	1959-08-09	m	site_a	199.2
## 23	S0023	A. Garcia	10/28/1974	M	site_a	184.7
## 24	S0024	B. Garcia	2011-12-30	M	siteB	169.5
## 25	S0025	A. Johnson	05/21/1972	f	Site C	143.1
## 26	S0026	J. KIM	30-Jan-2016	U	Site C	202.3
## 27	S0027	K. Patel	19-Apr-1991	m	Site C	111.0
## 28	S0028	S. Johnson	09-Nov-1985	M	SITE-A	102.1
## 29	S0029	T. Martinez	05.17.60	Female	Site C	148.5
## 30	S0030	R. Garcia	10-Dec-1976	U	Site C	123.2
## 31	S0031	J. Martinez	06/07/1986	m	Site B	211.4
## 32	S0032	J. Garcia	23-Jul-2009	F	Site C	112.3*
## 33	S0033	K. Martinez	07.14.56	U	siteB	105.0
## 34	S0034	B. Kim	06.07.12	M	Site C	91.8
## 35	S0035	K. Martinez	04.15.95	unknown	Site C	225.1
## 36	S0036	K. Patel	13-Dec-1956	Female	SITE-A	230.7
## 37	S0037	B. Kim	09.06.98		Site C	145.6
## 38	S0038	R. Nguyen	1987-08-05	unknown	Site B	157.4
## 39	S0039	K. Lee	12/02/1997	U	Site C	74.9
## 40	S0040	M. Brown	04.17.18	m	Site C	145.9
## 41	S0041	M. Garcia	06/21/2008	M	site_a	221.1
## 42	S0042	K. Nguyen	01/20/1965	Male	Site A	174.2
## 43	S0043	M. Garcia	2006-04-10	F	SITE-A	223.1*
## 44	S0044	R. Garcia	08/12/1980	U	Site A	148.0

## 45	S0045	S. Martinez	04.13.98	unknown	Site C	176.6
## 46	S0046	M. Brown	12.12.09		Site C	128.5
## 47	S0047	S. DAVIS	02/21/1967	Female	Site B	169.9
## 48	S0048	B. Nguyen	2012-01-24	m	SITE-A	155.2
## 49	S0049	S. Nguyen	1960-10-28	f	siteB	241.2
## 50	S0050	S. Garcia	1976-07-30	f	Site A	197.4
## 51	S0051	J. Brown	06-Feb-1951	unknown	SITE-A	103.2
## 52	S0052	K. Garcia	08/02/2011	Male	siteB	240.8
## 53	S0053	M. Davis	20-Sep-1989	U	siteB	80.8
## 54	S0054	J. Patel	1956-09-25	M	Site C	176.9
## 55	S0055	B. Patel	2006-01-28	U	site_a	186.6
## 56	S0056	S. Brown	14-Oct-2011	U	siteB	N/A
## 57	S0057	R. Martinez	1958-03-12	M	Site C	217.9
## 58	S0058	S. Lee	02.08.89	F	Site B	130.0
## 59	S0059	T. DAVIS	25-Nov-1957	m	Site C	142.8
## 60	S0060	R. Martinez	02.09.06	M	siteB	93.6
##	glucose_unit		notes			
## 1	mg/dl					
## 2	mg/dl					
## 3	MG/DL					
## 4	MG/DL					
## 5	mg/dl					
## 6	mmol/L					
## 7	mg/dl					
## 8	MG/DL					
## 9	mg/dl					
## 10	mg/dl		re-draw requested			
## 11	mmol/L					
## 12	mg/dl		re-draw requested			
## 13	mg/dL					
## 14	mmol/L					
## 15	mmol/L					
## 16	mmol/L					
## 17	mmol/L					
## 18	mg/dl		re-draw requested			
## 19	MG/DL					
## 20	mg/dL					
## 21	mg/dL					
## 22	mg/dl					
## 23	MG/DL					
## 24	mmol/L					
## 25	mg/dl					
## 26	mg/dl					
## 27	mg/dL					
## 28	MG/DL		re-draw requested			
## 29	mg/dl					
## 30	mg/dl					
## 31	MG/DL					
## 32	MG/DL					
## 33	mmol/L					
## 34	mg/dl					
## 35	mmol/L					
## 36	mg/dL					
## 37	mg/dL					

```
## 38      mmol/L
## 39      mmol/L
## 40      mg/dL
## 41      mg/dl
## 42      MG/DL
## 43      mg/dL
## 44      MG/DL
## 45      MG/DL
## 46      mmol/L
## 47      mg/dl
## 48      MG/DL
## 49      mg/dl
## 50      mg/dl
## 51      mg/dL
## 52      mg/dL
## 53      mg/dL
## 54      MG/DL re-draw requested
## 55      MG/DL
## 56      mmol/L
## 57      mg/dL
## 58      mg/dL
## 59      mg/dL
## 60      MG/DL
```

```
# =====

# Date of Birth Standardization

# Standardization 2-digit years
dot_dates <- str_detect(messy_sample$dob, "\\.")

messy_sample$dob[dot_dates] <- paste0(
  ifelse(
    as.numeric(str_sub(messy_sample$dob[dot_dates], 7, 8)) <= 25,
    "20",
    "19"
  ),
  str_sub(messy_sample$dob[dot_dates], 7, 8),
  ".",
  str_sub(messy_sample$dob[dot_dates], 1, 5)
)

# Standardization date formats
messy_sample$dob <- case_when(
  str_detect(messy_sample$dob, "/") ~
    as.Date(messy_sample$dob, "%m/%d/%Y"),

  str_detect(messy_sample$dob, "\\.") ~
    as.Date(messy_sample$dob, "%Y.%m.%d"),

  str_detect(messy_sample$dob, "^\\d{4}-") ~
    as.Date(messy_sample$dob, "%Y-%m-%d"),

  TRUE ~
```

```

    as.Date(messy_sample$dob, "%d-%b-%Y")
)

messy_sample$dob <- format(messy_sample$dob, "%Y/%m/%d")

messy_sample$dob

## [1] "1962/07/09" "1974/12/16" "1953/11/24" "1975/12/18" "1997/12/02"
## [6] "1955/12/09" "1969/09/04" "1997/06/21" "2018/05/11" "1951/10/01"
## [11] "1997/10/16" "2008/04/21" "2006/08/14" "1986/07/09" "1980/06/10"
## [16] "1998/07/01" "1973/10/17" "2002/05/30" "2005/07/26" "2017/06/15"
## [21] "1964/02/26" "1959/08/09" "1974/10/28" "2011/12/30" "1972/05/21"
## [26] "2016/01/30" "1991/04/19" "1985/11/09" "1960/05/17" "1976/12/10"
## [31] "1986/06/07" "2009/07/23" "1956/07/14" "2012/06/07" "1995/04/15"
## [36] "1956/12/13" "1998/09/06" "1987/08/05" "1997/12/02" "2018/04/17"
## [41] "2008/06/21" "1965/01/20" "2006/04/10" "1980/08/12" "1998/04/13"
## [46] "2009/12/12" "1967/02/21" "2012/01/24" "1960/10/28" "1976/07/30"
## [51] "1951/02/06" "2011/08/02" "1989/09/20" "1956/09/25" "2006/01/28"
## [56] "2011/10/14" "1958/03/12" "1989/02/08" "1957/11/25" "2006/02/09"

```

```

# =====

```

```

#Sex Standardization

```

```

sex_mess <- messy_sample$sex

```

```

sex_clean <- str_to_lower(sex_mess)

```

```

sex_clean <- case_when(
  sex_clean == "male" ~ "M",
  sex_clean == "m" ~ "M",
  sex_clean == "female" ~ "F",
  sex_clean == "f" ~ "F",
  sex_clean == "unknown" ~ "NA",
  sex_clean == "u" ~ "NA",
  TRUE ~ "NA"
)

```

```

messy_sample$sex <- sex_clean

```

```

messy_sample$sex

```

```

## [1] "F" "M" "F" "F" "M" "F" "NA" "M" "F" "F" "M" "NA" "NA" "F" "F"
## [16] "F" "M" "NA" "NA" "M" "M" "M" "M" "M" "F" "NA" "M" "M" "F" "NA"
## [31] "M" "F" "NA" "M" "NA" "F" "NA" "NA" "NA" "M" "M" "M" "F" "NA" "NA"
## [46] "NA" "F" "M" "F" "F" "NA" "M" "NA" "M" "NA" "NA" "M" "F" "M" "M"

```

```

# =====

```

```

# Enrollment Sites Standardization

```

```

site_clean <- str_to_lower(messy_sample$enrollment_site)

```

```

site_clean <- str_extract(site_clean, "[abc]")

```

```

site_clean <- str_to_upper(site_clean)

```

```

messy_sample$enrollment_site <- site_clean

# =====

#Glucose Value & Unit Standardization
glucose_clean <- str_remove(messy_sample$glucose_value, "\\*")
glucose_clean[glucose_clean == "NA"] <- "NA"
messy_sample$glucose_Value <- glucose_clean

glucose_clean <- as.numeric(glucose_clean)

## Warning: NAs introduced by coercion

mmol <- messy_sample$glucose_unit == "mmol/L"
glucose_clean[mmol] <- glucose_clean[mmol]*18

messy_sample$glucose_value <- glucose_clean
messy_sample$glucose_unit <- "mg/dl"

# =====

# Cleaned & Standardization sample

cleaned_sample <- messy_sample

View(cleaned_sample)

write.csv(cleaned_sample, "clean_sample.csv", row.names = FALSE)

```